Analysis of viral genomes using the Galaxy platform

Exercise 3: Illumina Assembly

Title	Galaxy
Training dataset:	**PRJEB43037** - In August 2020, an outbreak of **West Nile Virus** affected 71 people with meningoencephalitis in Andalusia and 6 more people in Extremadura (south-west of Spain), causing a total of eight deaths. The virus belonged to the lineage 1 and was relatively similar to previous outbreaks occurred in the Mediterranean region. Here, we present a detailed analysis of the outbreak, including an extensive phylogenetic study. This is one of the outbreak samples.
Questions:	• What is an assembly? • How can we evaluate our assembly?
Objectives:	• Understand the assembly concept. • Learn how to interpret assembly quality control metrics.
Estimated time:	40 min

1. Description

Sometimes, we don't have a reference genome to map our reads against, or we want to reconstruct a genome without any possible bias caused by a reference. In such cases, we need to perform ***de novo* assembly**. This type of analysis tries to reconstruct the original genome without any templates, using only the reads themselves. Some considerations to take into account beforehand:

- When we assemble, the longer the reads are and the longer the size of the library fragments, the easier it is for the assembler. That's why PacBio or Nanopore are recommended for assembly. Think of it like a puzzle: the bigger the pieces, the easier it is to form the image.
- It's almost imposible to reconstruct the entire genome of a large-genome microorganism with only one sequencing, although it can be done for smaller ones, like viruses.
- Assembly is not recommended for amplicon based libraries due to the depth of coverage unevenness and the amplicons' intrinsic bias.

2. Upload data to Galaxy

Training dataset

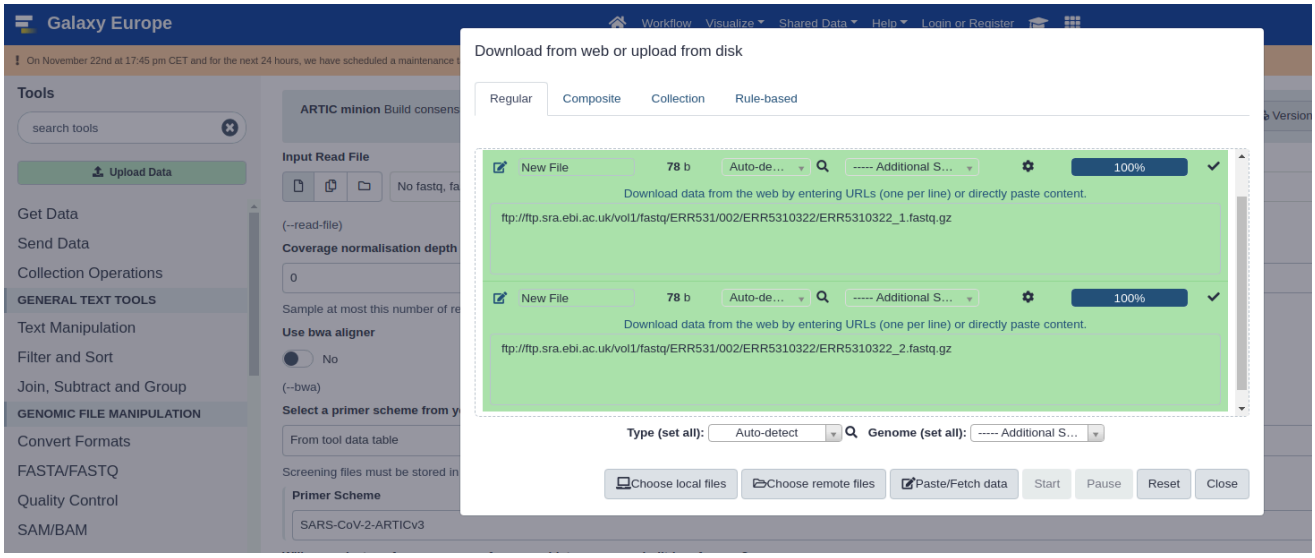
- **Experiment info**: PRJEB43037, WGS, Illumina iSeq 100, paired-end.
- **Fastq R1**: ERR5310322_1 - URL : `ftp://ftp.sra.ebi.ac.uk/vol1/fastq/ERR531/002/ERR5310322/ERR5310322_1.fastq.gz`
- **Fastq R2**: ERR5310322_2 URL : `ftp://ftp.sra.ebi.ac.uk/vol1/fastq/ERR531/002/ERR5310322/ERR5310322_2.fastq.gz`
- **Reference genome NC_009942.1**: FASTA -- GFF

Create new history

- Click on the + icon at the top of the history panel and create a new history with the name `Illumina Assembly` as explained **here**.

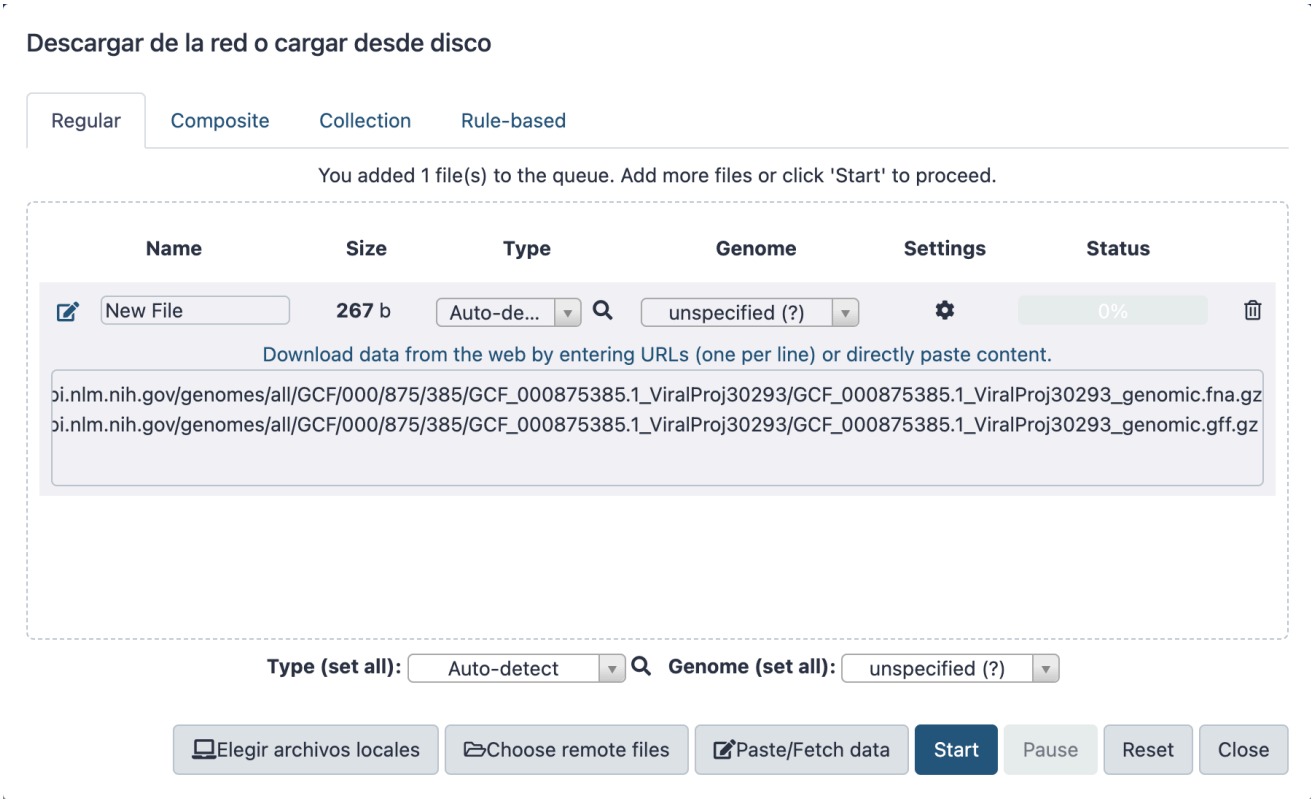
Upload data

- Import the read files `ERR5310322_1.fastq.gz` and `ERR5310322_2.fastq.gz`
    1. Click on **Upload**.
    2. Click on **Paste/Fetch data**.
    3. Copy URL for FASTQ R1 (select and Ctrl+C) and paste (Ctrl+V).
    4. Click on **Start**.
    5. Wait until the job finishes (green in history).
    6. Do the same for FASTQ R2.

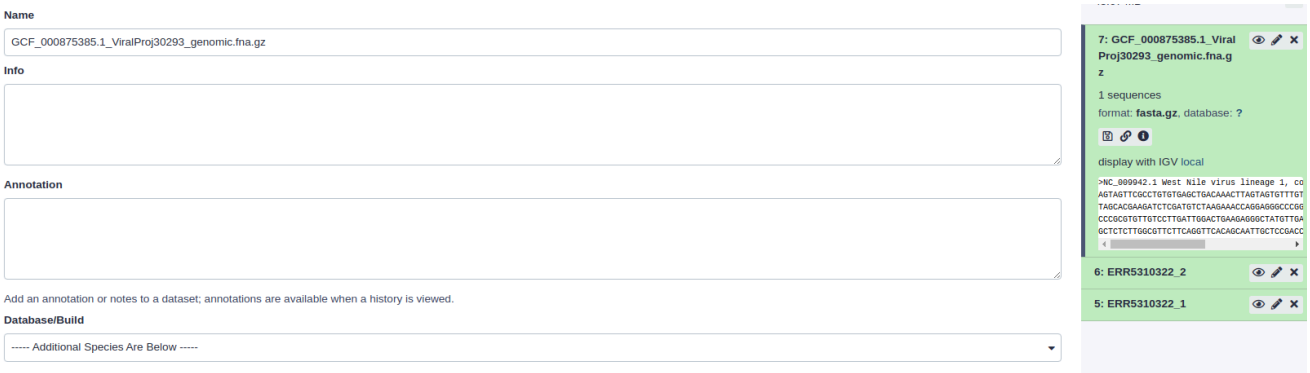


- You can then **add tags** (#R1 & #R2) to your reads, as we did in the preprocessing exercise.
- Import the **reference genome** and **GFF files** that will be used for assembly quality assessment at the end of the practice. You can add tags to them as well.

```
https://ftp.ncbi.nlm.nih.gov/genomes/all/GCF/000/875/385/GCF_000875385.1_ViralPro
https://ftp.ncbi.nlm.nih.gov/genomes/all/GCF/000/875/385/GCF_000875385.1_ViralPro
```



- You can rename the reference genome and GFF files if you like:
    1. Click on the ✏️ for the reference file in the history.
    2. Change the name to `NC_009942.1`.



3. Assemble reads with SPAdes

1. Search SPAdes in the search tool box and select **rnaviralSPAdes - de novo assembler for transcriptomes, metatranscriptomes and metaviromes**.

> Note

SPAdes is an assembler thas has multiple modes depending on the nature of the reads to be used for assembly. For instance, it has:

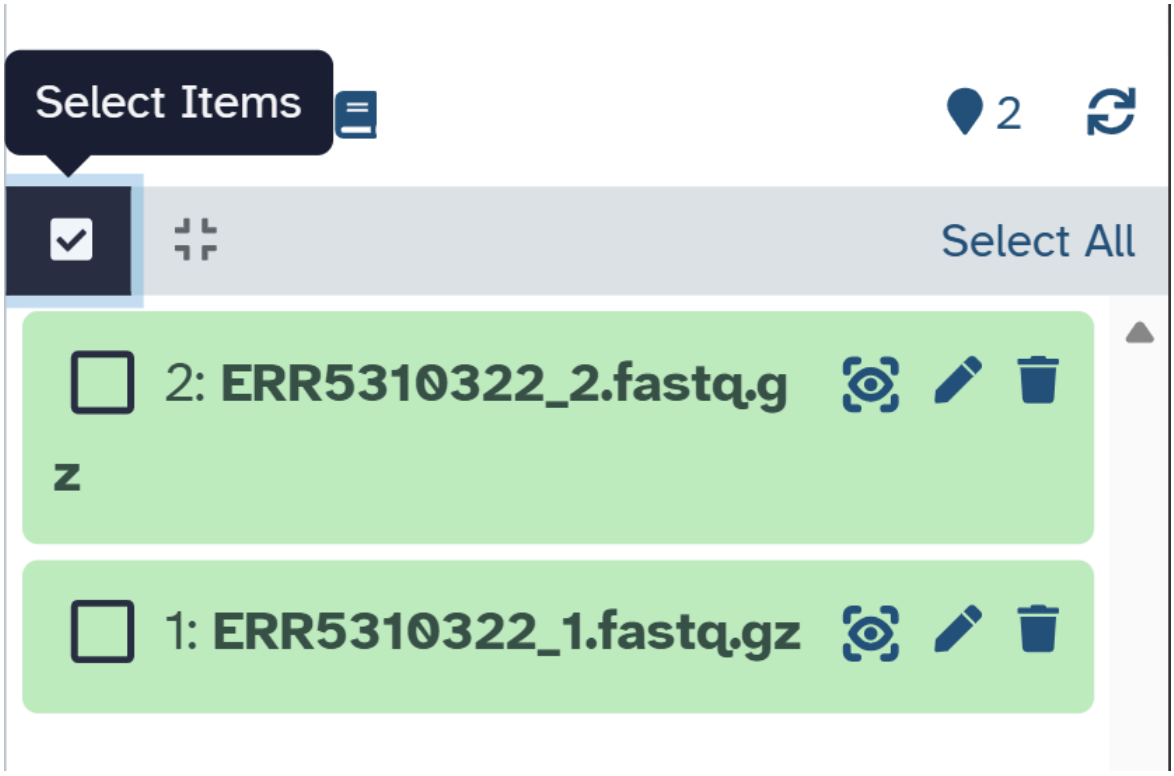
- metaSPAdes: assembly of metagenomic data containing sequences from multiple organisms.
- rnaSPAdes: transcriptome assembly from RNA-seq data.
- rnaviralSPAdes: assembly of RNA viral genomes from viral RNA-seq data.
- plasmidSPAdes and metaplasmidSPAdes: recovery of plasmids from whole-genome and metagenomic datasets, respectively.
- metaviralSPAdes: recovery of viral sequences from metagenomic data.

2. Single-end or paired-end short-reads > **Paired-end: list of dataset pairs**.

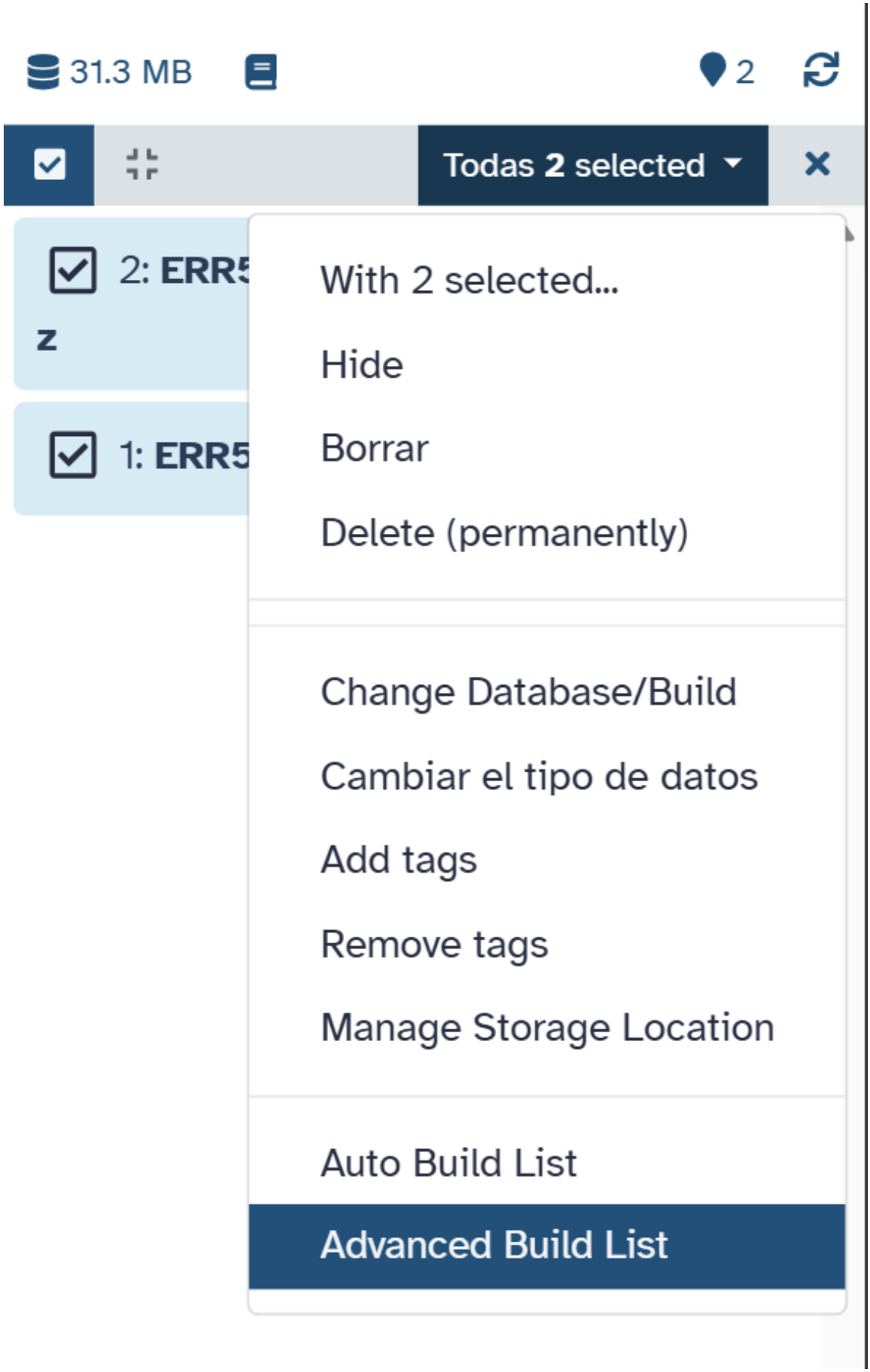
⚠ Warning

To be able to continue, it is necessary to first create a **list of paired datasets**, given our FASTQ files, which will then be used for the rest of the steps of this workshop. In order to do so, we must:

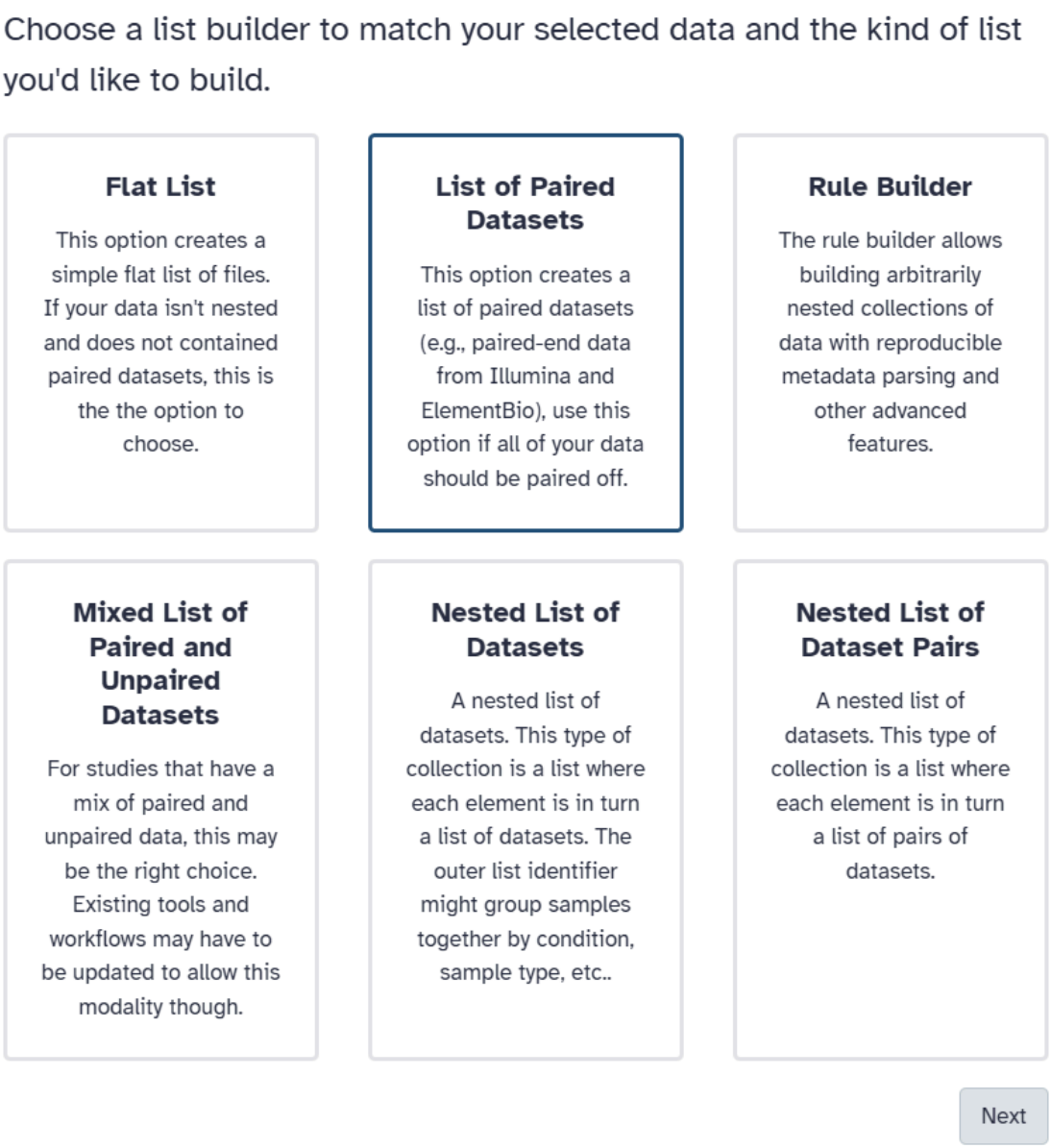
1. **Select our fastqsanger.gz files that we uploaded in the beginning, and then click on 2 of X selected**.



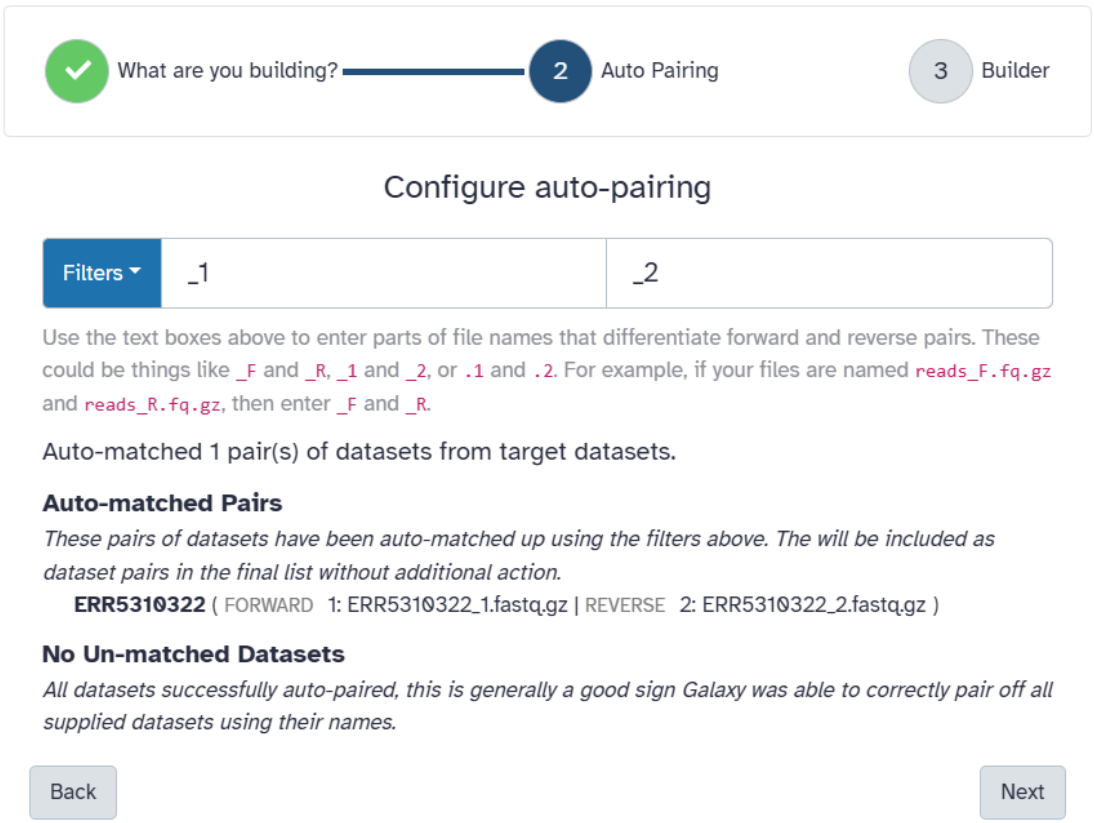
2. **Select Advanced Build List**.



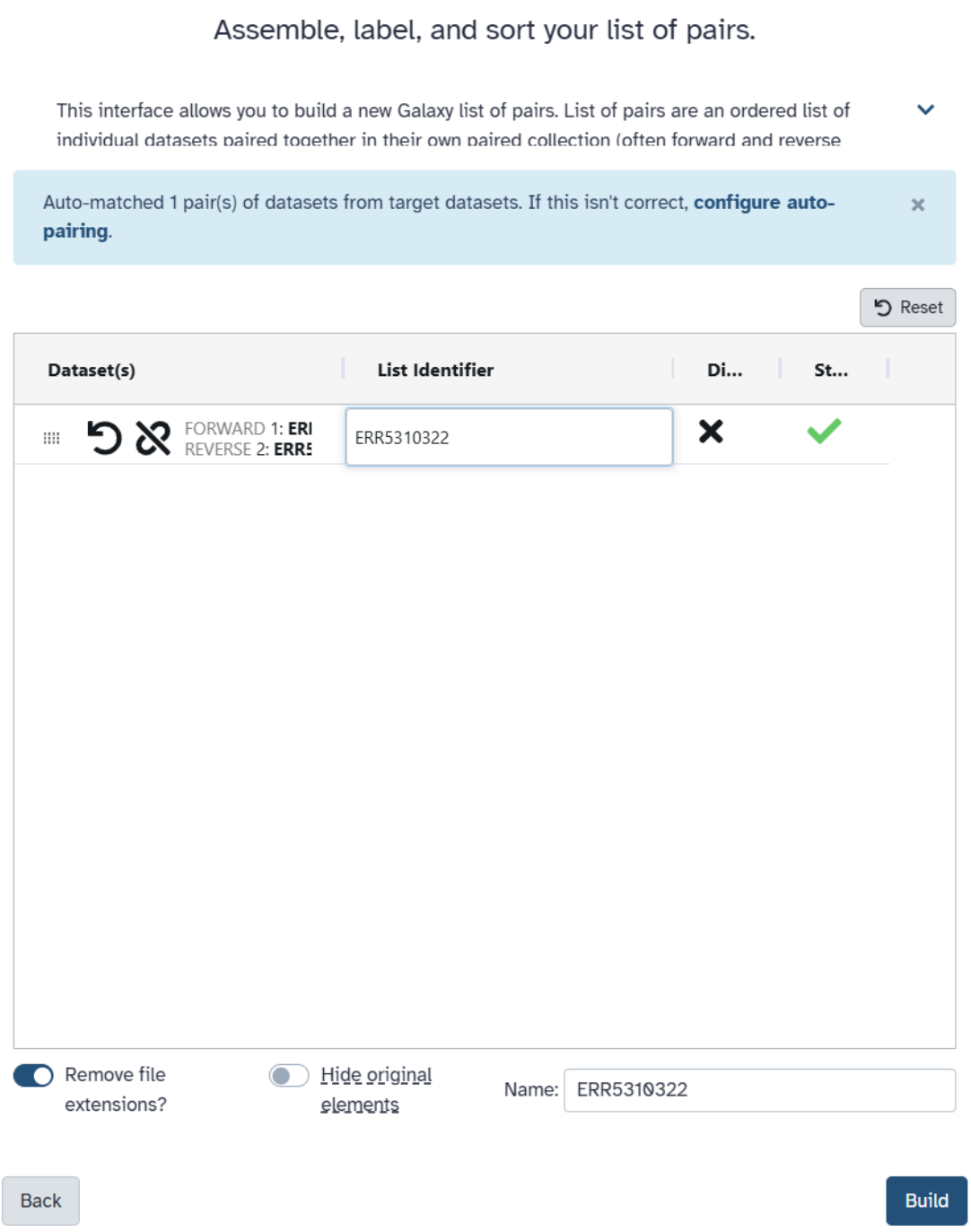
3. **Select the List of Paired Datasets box and click on *Next*.**



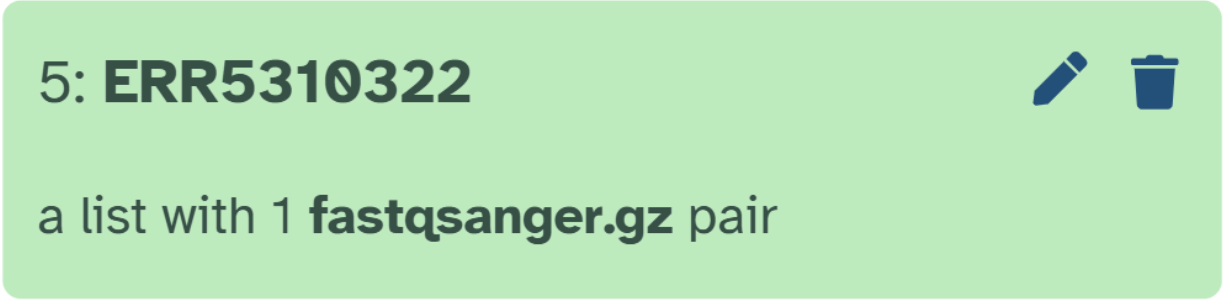
4. **Check the auto-pairing configuration and click again on *Next*.**



5. **Give a name to your list and finally click on *Build***.



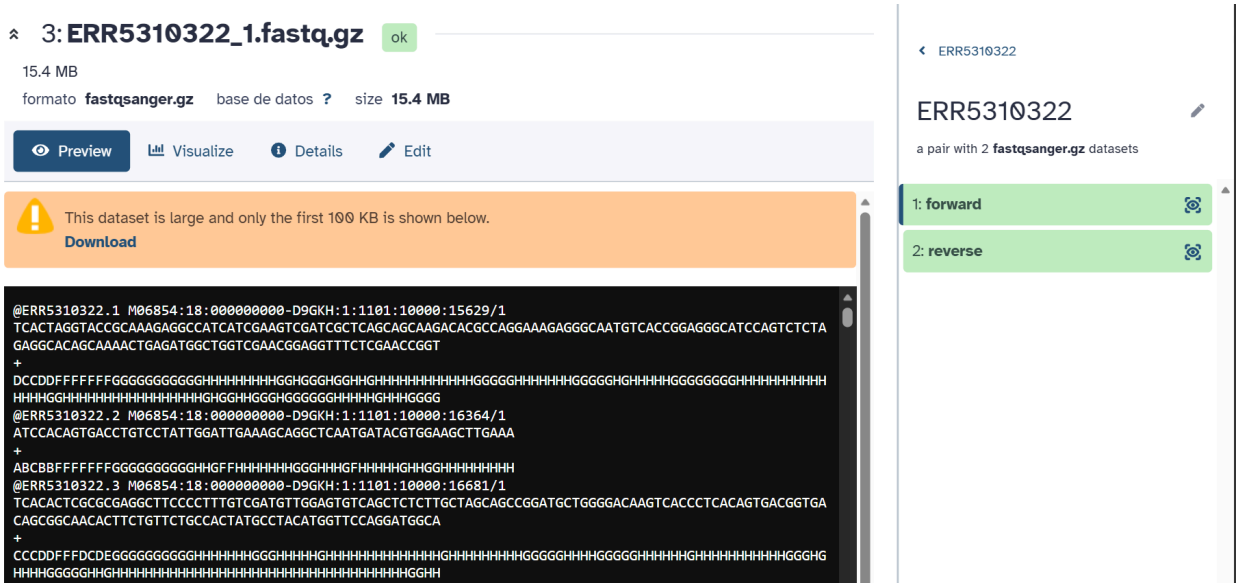
After having done this, a list of datasets will have been generated:

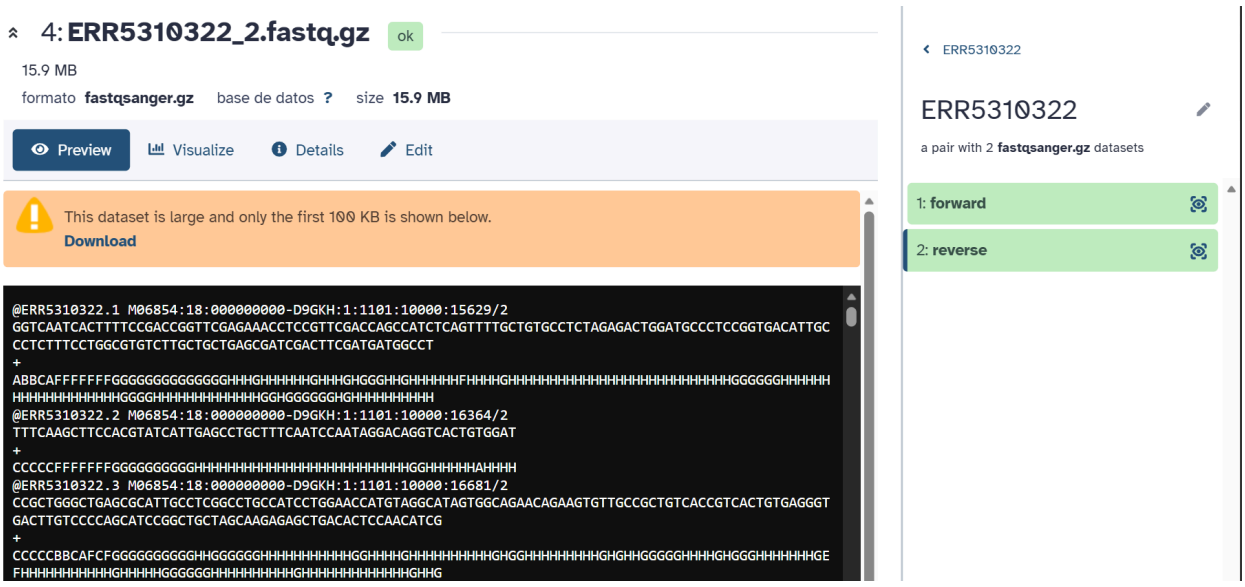


1: **ERR5310322**

a pair with 2 **fastqsanger.gz** datasets

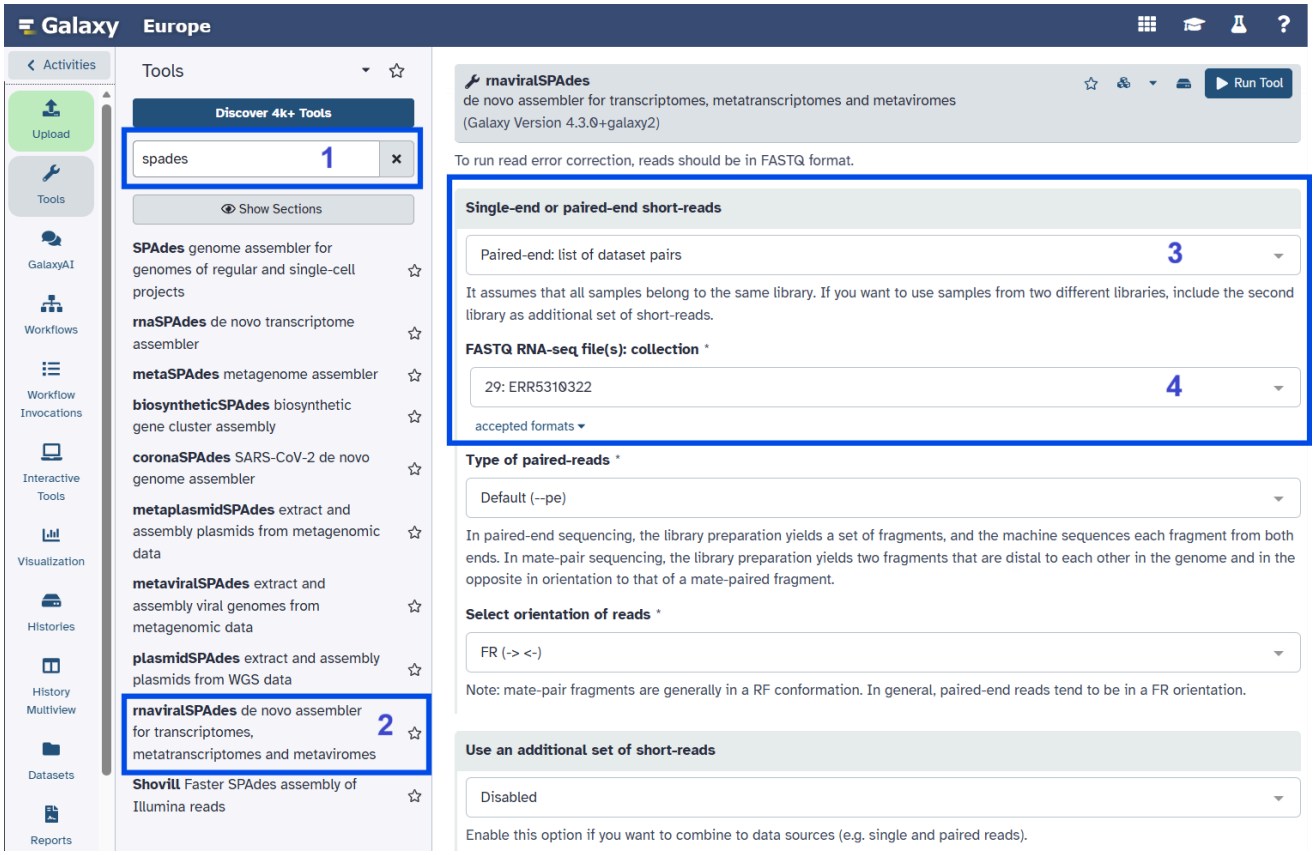
This list will have the forward and reverse reads:

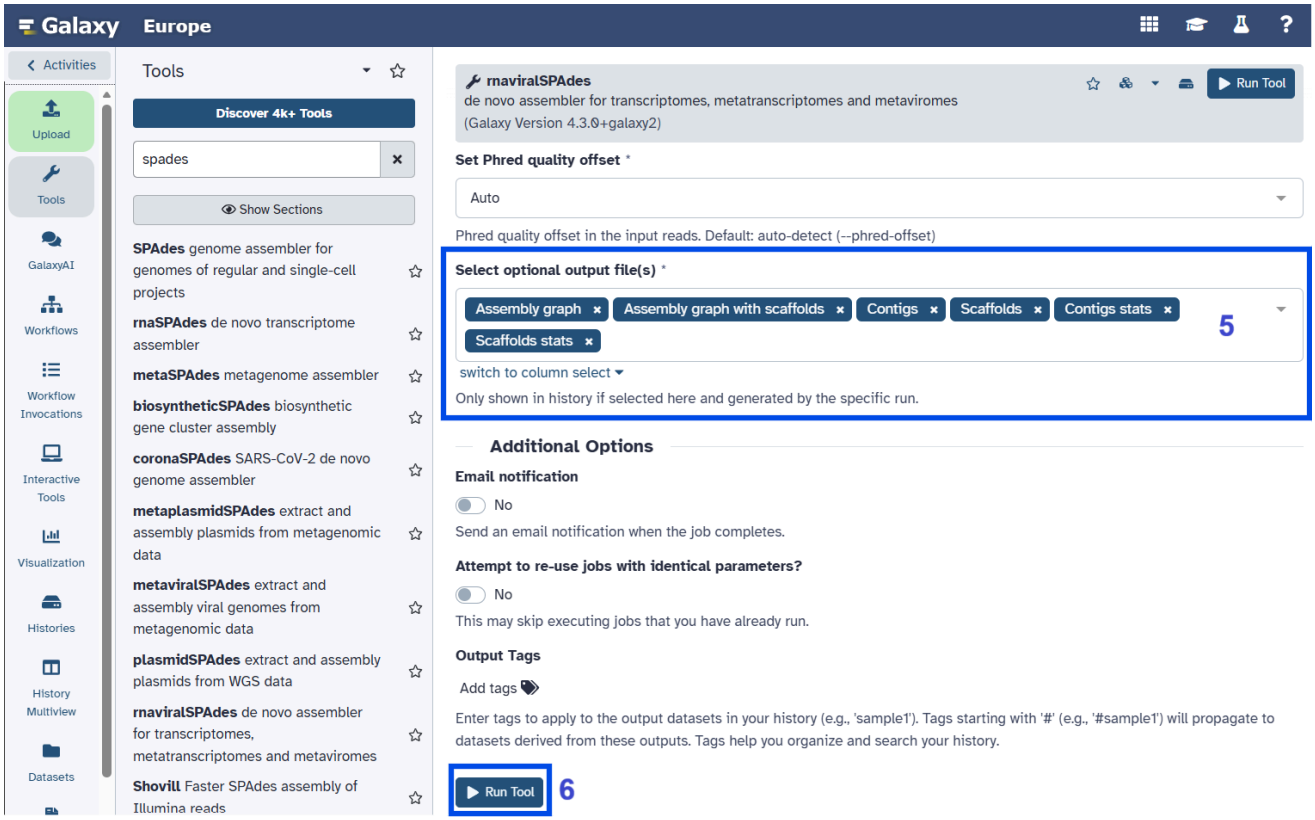




Once this is done, you'll be able to select your list as input of SPAdes.

1. FASTQ RNA-seq file(s): collection: **ERR5310322**
2. Select optional output file(s) > **Scaffolds stats** and **Contigs stats**, along with the options selected by default.
3. Click on **Run Tool** and wait.





> ⚠ Warning
>
> **Assembly takes time!** There is no such thing as Assembly in real time. This process might take long.

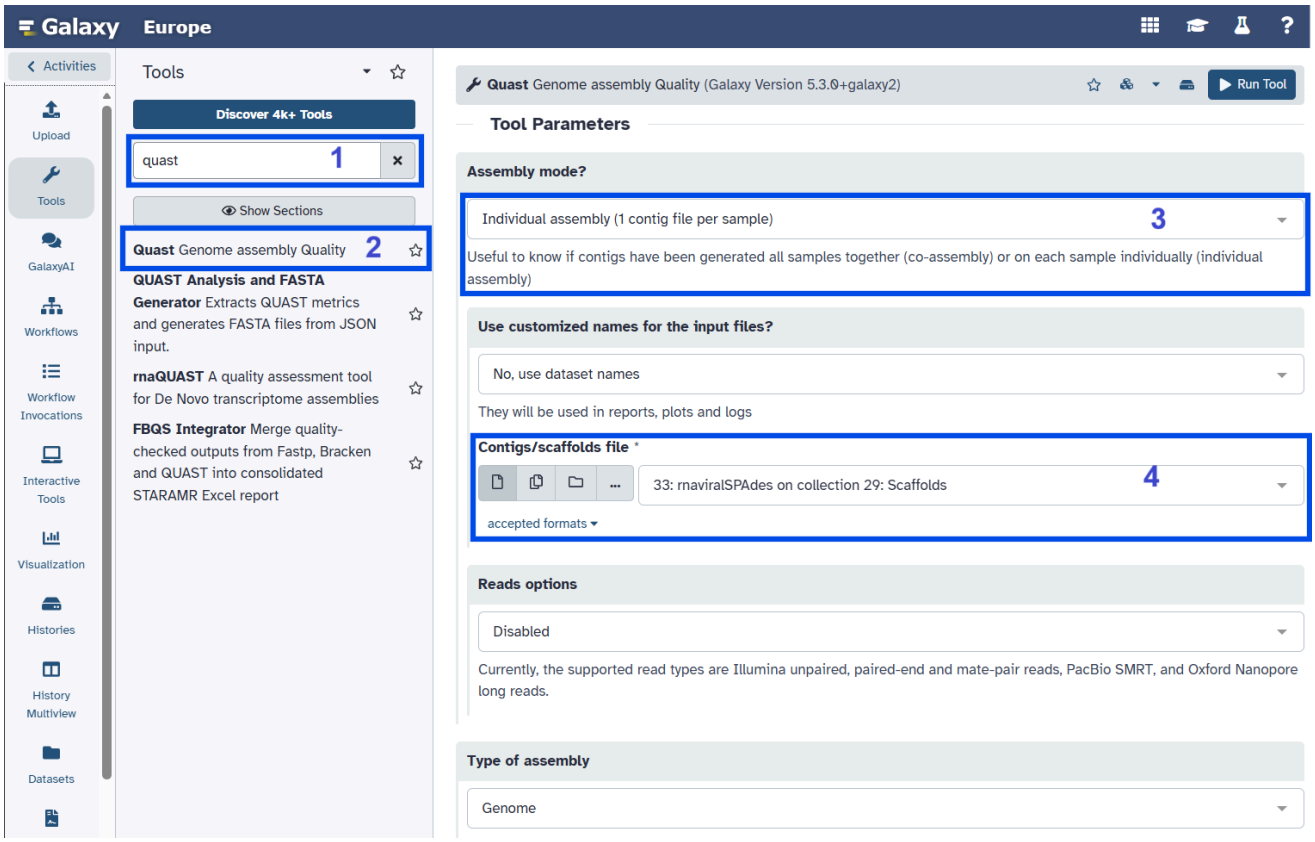
Questions:

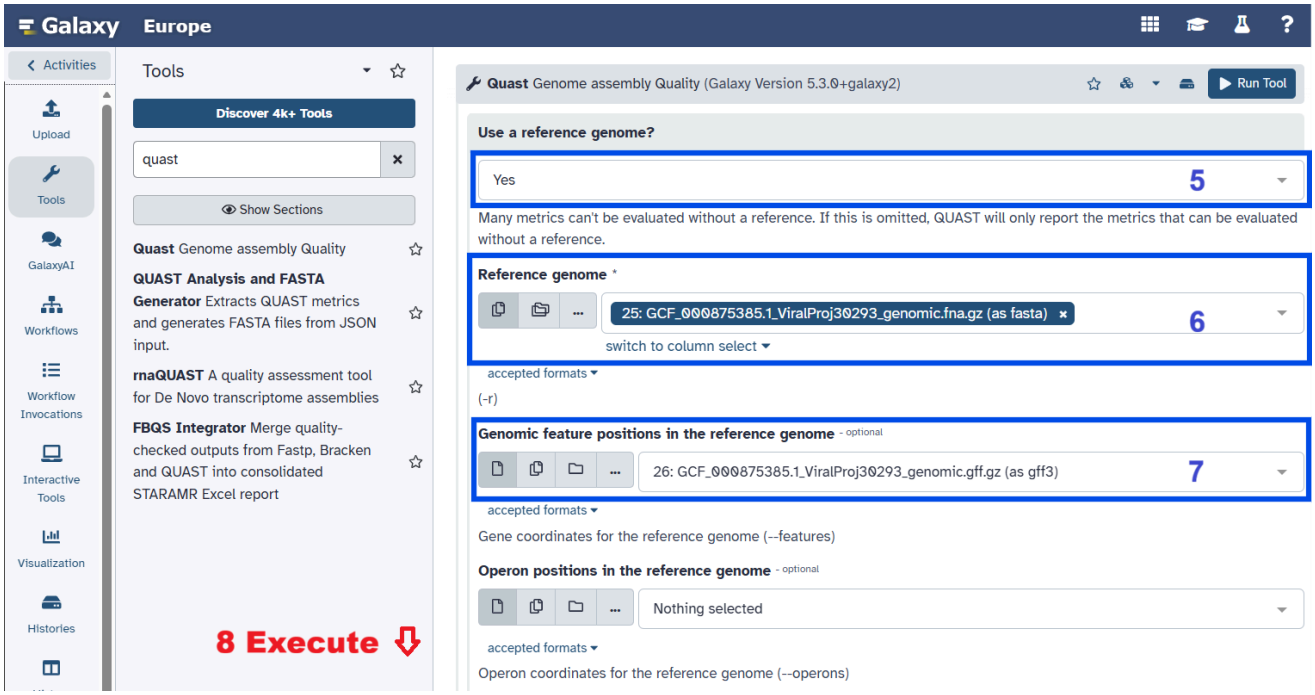
Click on the **eye icon** in the history: **Spades Contigs stats**.

▶ How many contigs have been assembled?

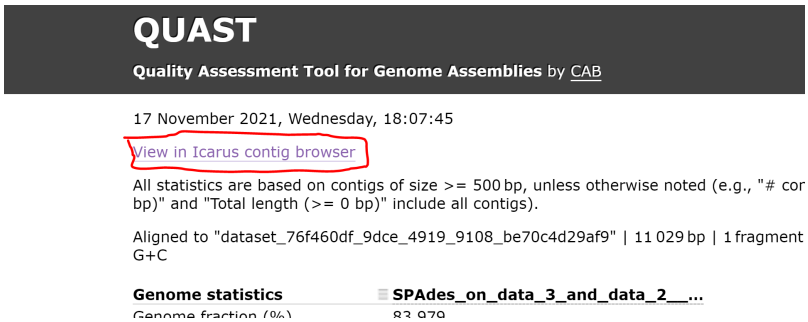
4. Assembly quality control with QUAST

1. Search **QUAST** in the search tool box and select **Quast - Genome assembly Quality**.
2. **Assembly mode? > Individual assembly (1 contig file per sample)**.
3. Contigs/scaffolds file > **rnaviralSpades Scaffolds**
4. Use a reference genome? > **Yes**. Select the **FASTA file** previously loaded.
5. Genomic feature positions in the reference genome > **GFF file** previously loaded.





6. **Run Tool**.
7. Click on the eye icon of the **QUAST HTML report**.
   - ▶ How much of our reference genome have we reconstructed?
   - ▶ How many contigs do we have greater than 1000 pb?
   - ▶ How long is the largest contig in the assembly?
   - ▶ Which is the N50?
8. Open the **Icarus** viewer in the QUAST report.



- ▶ How did the contig align against our reference genome?

> **This training history is available at: https://usegalaxy.eu/u/s.varona/h/illumina-assembly-101-tutorial**